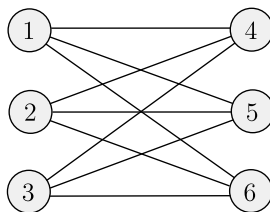


0.9 Write a formal description of the following graph.



## PROBLEMS

- 0.10 Find the error in the following proof that  $2 = 1$ .  
 Consider the equation  $a = b$ . Multiply both sides by  $a$  to obtain  $a^2 = ab$ . Subtract  $b^2$  from both sides to get  $a^2 - b^2 = ab - b^2$ . Now factor each side,  $(a + b)(a - b) = b(a - b)$ , and divide each side by  $(a - b)$  to get  $a + b = b$ . Finally, let  $a$  and  $b$  equal 1, which shows that  $2 = 1$ .
- 0.11 Let  $S(n) = 1 + 2 + \cdots + n$  be the sum of the first  $n$  natural numbers and let  $C(n) = 1^3 + 2^3 + \cdots + n^3$  be the sum of the first  $n$  cubes. Prove the following equalities by induction on  $n$ , to arrive at the curious conclusion that  $C(n) = S^2(n)$  for every  $n$ .
- $S(n) = \frac{1}{2}n(n + 1)$ .
  - $C(n) = \frac{1}{4}(n^4 + 2n^3 + n^2) = \frac{1}{4}n^2(n + 1)^2$ .
- 0.12 Find the error in the following proof that all horses are the same color.  
 CLAIM: In any set of  $h$  horses, all horses are the same color.  
 PROOF: By induction on  $h$ .
- Basis:** For  $h = 1$ . In any set containing just one horse, all horses clearly are the same color.
- Induction step:** For  $k \geq 1$ , assume that the claim is true for  $h = k$  and prove that it is true for  $h = k + 1$ . Take any set  $H$  of  $k + 1$  horses. We show that all the horses in this set are the same color. Remove one horse from this set to obtain the set  $H_1$  with just  $k$  horses. By the induction hypothesis, all the horses in  $H_1$  are the same color. Now replace the removed horse and remove a different one to obtain the set  $H_2$ . By the same argument, all the horses in  $H_2$  are the same color. Therefore, all the horses in  $H$  must be the same color, and the proof is complete.
- 0.13 Show that every graph with two or more nodes contains two nodes that have equal degrees.